# Operation Scenario and Vacuum Fields

VAFT Introductory Tutorial | Session 02

# Why this session exists

- A discharge does not simply happen — it is started.
- Read one backwards: from what the plasma did, to the field that let it begin.

# Three phases, three fingerprints

- **Breakdown** — avalanche, if the field lines are long enough.
- **Burn-through** — heat past the radiating impurities, or die.
- **Ramp-up** — the transformer drives the current.

# The vessel is part of the circuit

Vessel eddy currents are comparable to the plasma current and oppose the coils.

Every magnetic measurement sees the sum of all three.

# The vacuum field is a precondition

- $V_{\mathrm{loop}} = -\,\mathrm{d}\psi/\mathrm{d}t$ — the drive.
- $B_z$ — balances the hoop force.
- $n = -\dfrac{R}{B_z}\dfrac{\partial B_z}{\partial R}$ — decides whether that balance is *stable*: $0 < n < 1.5$.

# Session workflow

solve vessel currents $\rightarrow$ check against magnetics

$\rightarrow V_{\mathrm{loop}}$, $B_z$, decay index $\rightarrow$ breakdown time $\rightarrow$ the null

# Check the model before you use it

The vacuum-field comparison against measured magnetics comes *before* the physics derived from it, not after.